THE ACTION OF A RELATIVE CYCLE DEFINES A BIVARIANT CLASS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Gysin compatibility of the bivariant action (relative-cycles chapter, Lemma lemma-action-gysin, canonical tag 0H6E). cited at proof lines 2732-2733 and 2738-2739. Used both for factoring through rational equivalence and as one of the bivariant compatibilities.
- Factorization through rational equivalence (Chow Homology, Lemma chow-lemma-factors-through-rational-equivalence, canonical tag 0B7A). cited at proof lines 2733-2734. Turns the Gysin compatibility into factorization of the operation through rational equivalence.
- Bivariant class criterion from weaker data (Chow Homology, Lemma chow-lemma-bivariant-weaker, canonical tag 0F9A). cited at proof lines 2735-2736. Criterion: an operation on chow groups with the push-pull and base change compatibilities is a bivariant class.
- Push-pull compatibility of the action (relative-cycles chapter, Lemma lemma-action-push-pull, canonical tag 0H6B). cited at proof line 2738. One of the compatibilities feeding the bivariant-weaker criterion.
- Base change compatibility of the action (relative-cycles chapter, Lemma lemma-action-base-change, canonical tag 0H68). cited at proof lines 2738-2739. One of the compatibilities feeding the bivariant-weaker criterion.

Source reference. Original lemma 0H6E: sources/stacks-chow/tags.txt, source lines 21419–21419. Cited at proof lines 2732-2733 and 2738-2739: factorization and bivariance compatibility.

Source reference. Original lemma 0B7A: sources/stacks-chow/tags.txt, source lines 14098–14098. Cited at proof lines 2733-2734: factorization through rational equivalence.

Source reference. Original lemma 0F9A: sources/stacks-chow/tags.txt, source lines 19070–19070. Cited at proof lines 2735-2736: criterion concluding bivariance.

Source reference. Original lemma 0H6B: sources/stacks-chow/tags.txt, source lines 21418–21418. Cited at proof line 2738: one of the compatibilities for the criterion.

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Source reference. Original lemma 0H68: sources/stacks-chow/tags.txt, source lines 21413–21413. Cited at proof lines 2738-2739: one of the compatibilities for the criterion.

Source reference. Original section 0H65: sources/stacks-chow/tags.txt, source lines 21410–21410. Cited at statement line 2719: fixes the ambient data (S, delta) and the action.

Proposition 0.2. *Let (S, δ) be as in Section 0H65. Let $X \to Y$ be a morphism of schemes locally of finite type over S. Let $r \geq 0$ and let $\alpha \in z(X/Y, r)$ be a relative r-cycle on X/Y. The rule that to every morphism $g : Y' \to Y$ locally of finite type and every $e \in \mathbf{Z}$ associates the operation*

$$g^*\alpha \cap - : Z_e(Y') \to Z_{r+e}(X')$$

where $X' = Y' \times_Y X$ factors through rational equivalence to define a bivariant class $c(\alpha) \in A^{-r}(X \to Y)$.

Proof. The operation factors through rational equivalence by Lemma 0H6E and Chow Homology, Lemma 0B7A. The resulting operation on chow groups is a bivariant class by Chow Homology, Lemma 0F9A and Lemmas 0H6B, 0H68, and 0H6E. □